COURSE PROSPECTUS

Course Name: Training Workshop on Industrial Automation & Control

Duration: 60 Hours/ 2weeks.

Class Timing: 6 hours a day. Monday to Friday

Mode: Offline (Theory + Practical)

Course Fee: INR9,499.00/-

Course Description:

This course provides an overall exposure to the technology of Industrial Automation and Controls. This course discusses a wide range of related topics from the advantage and architecture of automation systems, components used in developing control systems like Programmable Logic Controllers (PLC), RTU, drives, HMI, sensors, transmitters, hydraulic, pneumatic, and electric actuators, various controlling schemes of actuators, industrial communications, DCS, SCADA systems and IoT solutions used for Industrial Automation. The participants will get exposure to configuration of Industrial Controllers and product selections. This course will help students to develop an in-depth knowledge of control systems and their functioning.

Who Can Attend?

- Students pursuing Engineering/Diploma/MSc/ITI with Electrical/EEE/Electronics/ Electronics and communication/ Instrumentation and Control/ Instrumentation/ Applied Electronics/Mechatronics/ Mechanical / Computer Science background.
- Any Interested student.
- Professionals working in Industry.
- Research scholars, Teaching professionals & faculties.

Course Outcome:

On successful completion of the course students will be having knowledge about

1. Different products and techniques used in industrial automation.
2. Types of field devices involved in automation and control systems and their configuration.
3. Types of PLC, selection of PLC and related modules.
4. PLC Programming, functions, and techniques in-built in PLC, using the programming software.

5. VFD and Servo drives. Selection and tuning of drives.
6. Configuration and integration of drives with PLC.
7. Programming of HMI and communication with PLC
8. SCADA and IoT systems

Module Description:

Sr. No.	MODULE TITLE	DURATION
1.	Introduction to Industrial Automation Systems	2
2.	Architecture of Industrial Automation Systems	2
3.	Introduction to Field Instruments – input & output devices	3
4.	Actuating devices and controlling of actuating devices	3
5.	Introduction to PLC based control systems	2
6.	PLC Hardware & PLC Programming	12
7.	Variable Frequency Drives	6
8.	Servo Drives	6
9.	HMI & HMI Programming	6
10.	SCADA Systems & IoT Solutions	3
11.	Higher Level Automation system and Communication Protocols	2
12.	Low Voltage Switchgear and Control Components	3
13.	Introduction to Control Panels & Designing	9
14.	Introduction to industrial robots	1
	TOTAL HOURS	**60**

Course Layout:

1. **Introduction to Industrial Automation Systems (1 hour)**
   - What is automation?
   - Historical overview and evolution.
   - Brief about components involved in Control and Automation systems.
   - Advantages and benefits of industrial automation.
   - Applications of control & automation systems in various industries.
2. **Architecture of Industrial Automation Systems**
   - Components involved in developing a control system.
   - System configuration and design parameters.

- System architecture development and adaptation.
- Industrial Voltage levels, Signals & Measurements.

3. **Introduction to field instruments – IO devices**
   - Types of input devices and signals.
   - Types of output devices.
   - Reference and feedback signals.
   - High speed input devices.
   - Sensors and transmitters.
   - Signal conditioners and amplifiers.
4. **Actuating Devices & Controlling of Actuating Devices**
   - Various output/ actuating devices used in industry.
   - Hydraulic and pneumatic systems.
   - Control Valves.
   - Induction motor, Stepper motor and Servo Motor
5. **Introduction to PLC based control systems (6 hours)**
   - Introduction to PLC
   - Types of PLC
   - Nomenclature and selection of PLCs
   - Application & Advantages
6. **PLC Hardware & PLC Programming (4 hours)**
   - CPU Module, extension module, IO Modules, adapters, and communication cards.
   - Introduction to PLC Programming
   - PLC programming languages and programming software.
   - Functions and techniques in programming.
   - PLC Wiring logics.
   - Sequential development of PLC program.
   - Application discussion
   - Practical implementation and development of programs for simple applications.
7. **Variable Frequency Drives**
   - Functioning of VFDs.
   - Speed control of induction motors using VFDs.
   - Selection and types of VFD based on application.
   - Parameters configuration and tuning of VFDs.
   - Practical implementation.
8. **Servo Drives (3 hours)**
   - Servo Motor and closed loop system.
   - Introduction to Servo amplifier and configuration software.
   - Selection and types of servo motors and drives.
   - Parameter configuration and tuning of servo amplifier.
   - Practical implementation.
9. **HMI & HMI Programming**
   - Introduction to HMI

- HMI Design Principles
- Screen development software.
- Integration and communication with PLC.
- Practical implementation.

10. SCADA Systems & IoT Solutions (3 hour)

- Introduction to Supervisory Control and Data Acquisition (SCADA)
- Difference Between HMI& SCADA
- Communication with control systems
- Historical Data handling
- Alarm management, Security & Access control.
- Data management, Dashboard development and ease of access.
- Practical Sessions.

11. Higher Level automation systems & communication protocols

- Distributed control systems
- Communication protocols
- Multiple levels of data transfer and communication.

12. Low Voltage Switchgear & Control Components

- Introduction to LVS
- Ratings and specifications
- Product selections and applications
- Components used in control panels.
- Real time understanding of products.

13. Introduction to control panels and designing.

- Designing of control panels
- Types of enclosures and protection.
- Practical wiring session of control panel.

14. Introduction to Industrial robots

- Industrial robots and types
- Application discussion.

Note:

- During the course period, students can visit our facility on Saturday to experiment with the automation components on their own.